



NUST
NATIONAL UNIVERSITY
OF SCIENCES & TECHNOLOGY

Assignment 2

Data Structures and Algorithms Software Engineering

September 29, 2026

CLO-2: Apply knowledge to solve real-world problems using suitable data structures and algorithms. Students will learn to analyze the problem requirements and select the best solution approach.

Contents

1 Introduction	2
2 Rules	2
2.1 Academic Integrity	2
2.2 Plagiarism	2
3 Scoring	3
4 Assignment Structure	3
4.1 Part 1: LeetCode Biweekly Contest 167 Participation (Contest Period)	3
4.2 Part 2: LLM-Assisted Solution	4
4.3 Part 3: Class Activity	5
5 Contest Details	5
6 Submission Guidelines	5

1 Introduction

Welcome to your DSA Assignment 2! This assignment is structured around the **LeetCode Biweekly Contest 193**, which will take place next weekend. This assignment is designed to evaluate your problem-solving abilities, your understanding of data structures and algorithms, and your ability to learn from alternative approaches.

The assignment is divided into **three parts**:

- **Part 1:** Participation in the LeetCode Biweekly Contest 193
- **Part 2:** LLM-Assisted Solution
- **Part 3:** Class Activity

Each part is designed to build upon your understanding and help you develop a deeper understanding of algorithmic problem solving and code optimization.

2 Rules

2.1 Academic Integrity

- **Original Work:** All contest submissions must be your own original work during the contest period.
- **Prohibitions During Contest:** During the live contest, you are **strictly prohibited** from:
 - Using AI tools (ChatGPT, Claude, Copilot, etc.)
 - Collaborating with other students
 - Searching for solutions online
 - Copying code from any external source
- **LeetCode Contest Rules:** You must adhere to all LeetCode contest rules as outlined on the contest page. Violations or disqualification by LeetCode (including for use of AI tools) will result in an automatic **score of zero**.
- **Exceptions:** The use of LLMs is **only permitted in Part 2** of the assignment, after the contest has ended.

2.2 Plagiarism

- Plagiarism in any form will not be tolerated during the contest portion (Part 1).
- Submissions will be monitored for plagiarism.
- Students found to be in violation of these rules will receive a score of 0 for the entire assignment, and additional disciplinary action may be taken in accordance with institutional policies.

3 Scoring

Your final assignment score will be based on the quality and completeness of all three parts of the assignment.

- **Grading:** You are not penalized for incorrect solutions during the contest. What matters is that you make a genuine attempt to solve the problems and can explain your thought process.
- **Part 1:** Graded based on participation, submission evidence (screenshots), and explanation of your approach.
- **Part 2:** Graded based on the quality of your analysis comparing your solution with the LLM's solution.
- **Part 3:** Graded based on your performance in the class activity (details to be announced).

It is essential to demonstrate understanding of the problems and data structures used, as both successful and unsuccessful attempts will contribute to your overall score.

4 Assignment Structure

4.1 Part 1: LeetCode Biweekly Contest 193 Participation (Contest Period)

Objective: Participate in the live LeetCode Biweekly Contest 193 and attempt to solve the problems using your own knowledge and skills.

Requirements:

1. **Register and Participate:** Create or use your existing LeetCode account to participate in the contest at the scheduled time.
2. **Attempt Problems:** You are required to **try attempting all problems** from the contest. Even if you cannot solve them, partial or incorrect attempts will still be graded.
3. **Screenshots:** For each problem you attempt:
 - Screenshot of your code/solution
 - Screenshot of your submission result (whether accepted or not)
 - Screenshot showing your submission timestamp to prove contest participation
4. **Documentation:** For each problem you attempt, write a brief explanation covering:
 - Your initial understanding of the problem
 - The approach you decided to take and why
 - Data structures and algorithms you used
 - Any challenges you faced during implementation
 - Time complexity of your solution
 - What you would do differently if you had more time

Important Notes:

- You do not need to solve all the problems correctly to get full marks for this part.
- Focus on demonstrating your problem-solving process and understanding.
- Make sure to save your code even if it doesn't pass all test cases.

4.2 Part 2: LLM-Assisted Solution

Objective: After the contest ends, use an LLM (Large Language Model) to generate solutions for the same problems and analyze the differences between your approach and the LLM's approach.

Requirements:

1. **Choose an LLM:** Select any LLM of your choice (ChatGPT, Claude, Gemini, etc.).
2. **Generate Solutions:** For each of the problems you attempted:
 - Provide the problem statement to the LLM
 - Ask it to generate an optimal solution
 - Save the LLM's response and solution code
 - You must also submit a **screenshot of the LLM-generated code submission**, tested on LeetCode after the contest.
3. **Comparative Analysis:** For each problem, write a brief comparison that includes:
 - **Approach:** How does the LLM's approach differ from yours?
 - **Code Quality:** Differences in code structure, readability, and style
 - **Optimizations:** Did the LLM use better data structures or algorithms?
 - **Complexity:** Compare time of both solutions
 - **Your Strengths:** Where your solution performed better or equally well
 - **Weaknesses Identified:** Mistakes or inefficiencies in your own approach
4. **Share LLM Chat Link:** Provide the shareable link to your conversation with the LLM for verification.

4.3 Part 3: Class Activity

Objective: Participate in an in-class activity that will test your understanding of concepts from the contest problems.

Details:

- The activity will be announced in class after the contest weekend. It will be held on the 14th of October.
- Attendance is mandatory for this part.

5 Contest Details

- **Date:** Saturday, 10th October 2026
- **Time:** 7:30 PM (Pakistan Standard Time)
- **Duration:** 1 hour 30 minutes
- **Contest Link:** <https://leetcode.com/contest/biweekly-contest-193/>

Note: You will need to register for the contest before the contest starts, after 7:30 pm, the registrations will be closed. Additionally, ensure that the rules listed at the contest website are read thoroughly.

6 Submission Guidelines

All students are required to submit a single PDF file containing the following:

- Your Name, CMS ID, Section, LeetCode Username, **LeetCode Profile Link**, and Email.
- For each problem, provide:
 - Screenshot of your submission
 - The logic behind your solution (briefly)
 - Any optimizations you used
 - Any failed attempts and why they failed
 - Screenshot proof of your final ranking in the contest
- For each LLM solution, include:
 - LLM used
 - Link to the LLM conversation
 - Comparative analysis
 - Screenshot of the LLM-generated code submission

Important: Incomplete or incorrect attempts will still be graded based on your reasoning and understanding. Submission deadline for this part will be 11 October 11:59 pm.

Good Luck and Happy Coding! :)

Grading Criteria (Total 10 Marks)

Component	Marks
Participation in the LeetCode Contest	2
Self Attempt of the Problems	3
Analysis of the LLM Generated Code	2
In-Class Activity	3
Total	10